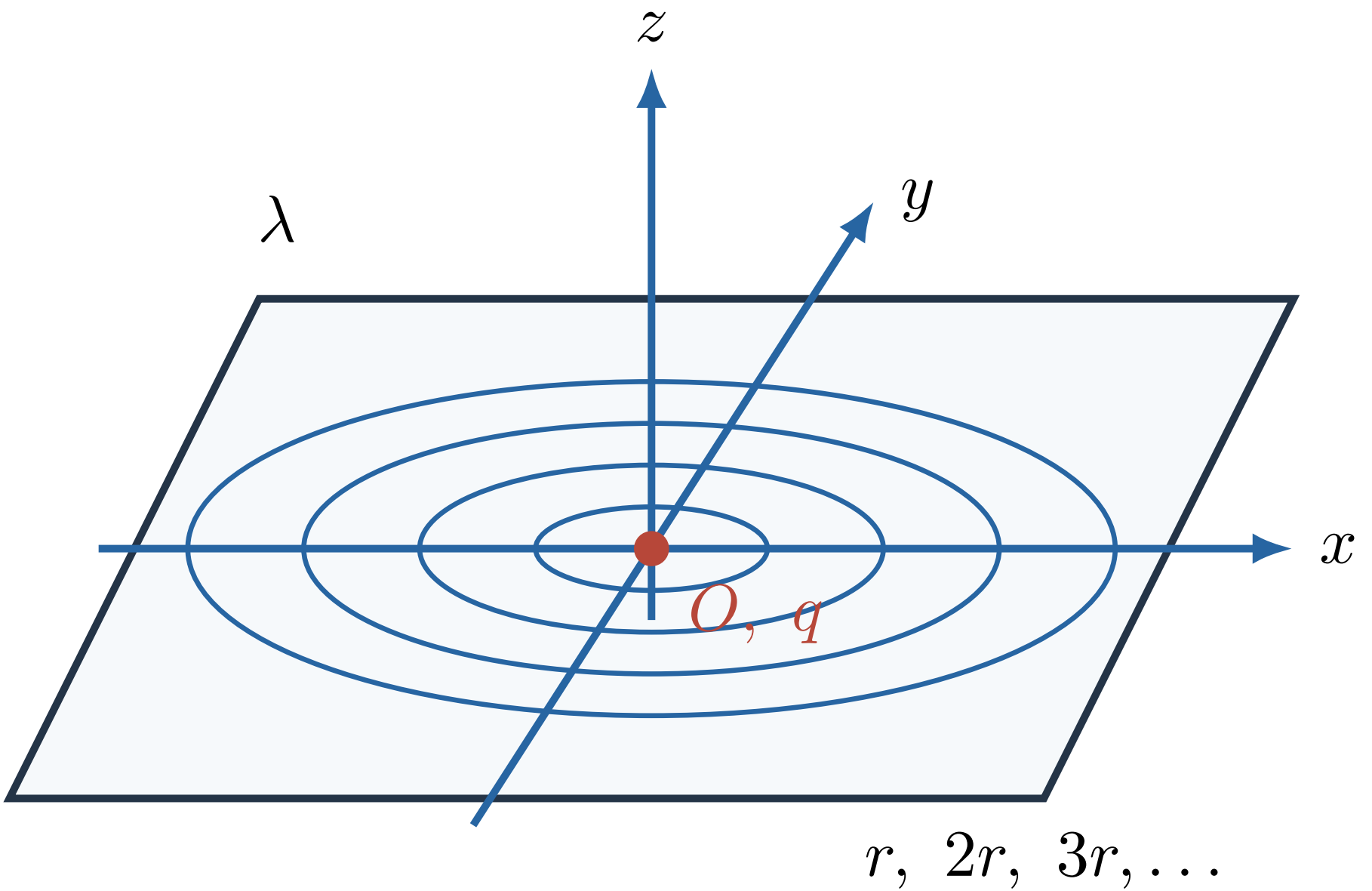
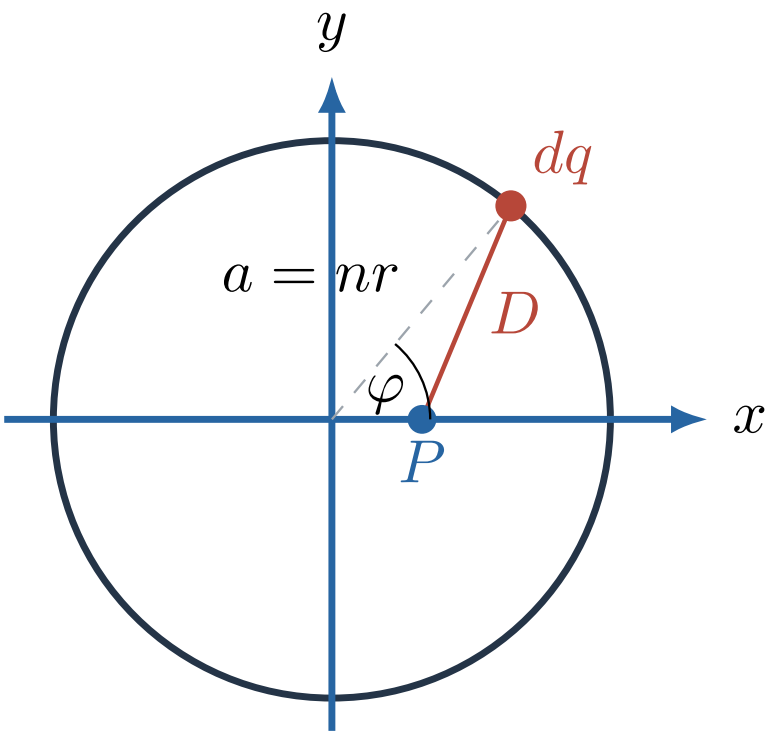
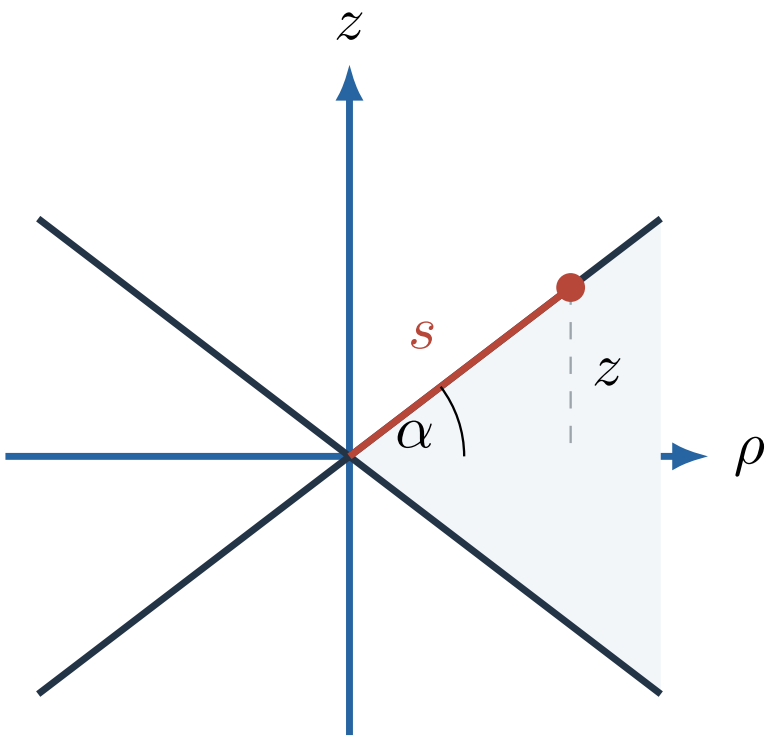
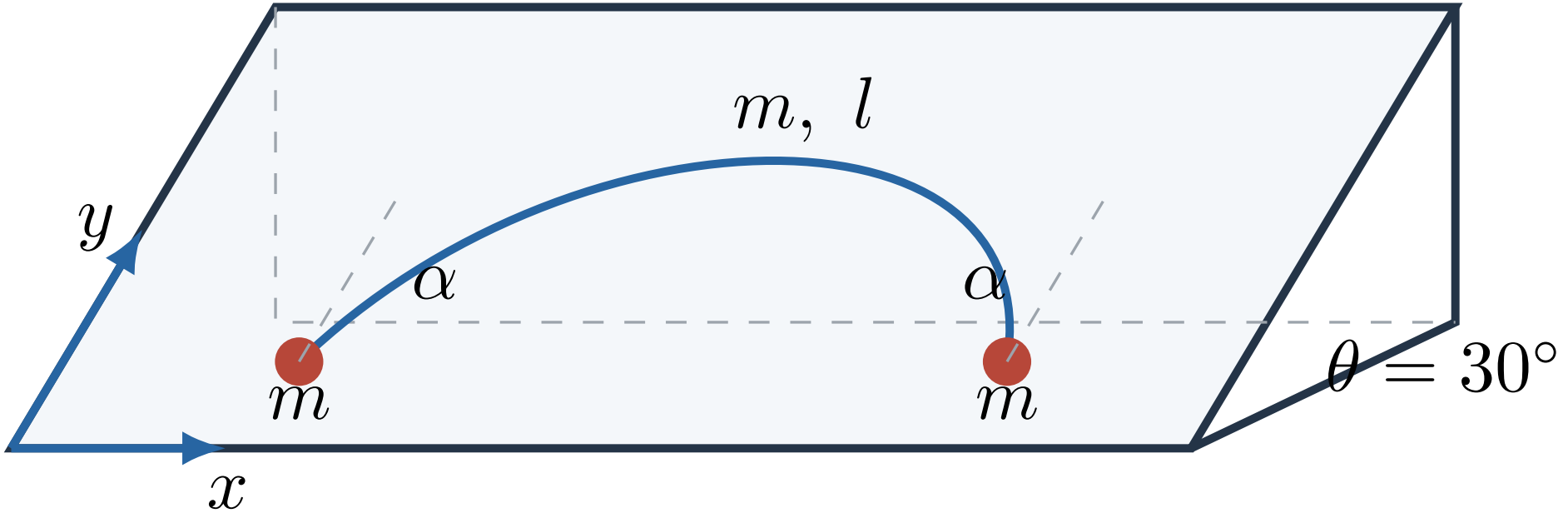

z
y
λ
x
O, q
r, 2r, 3r, . . .

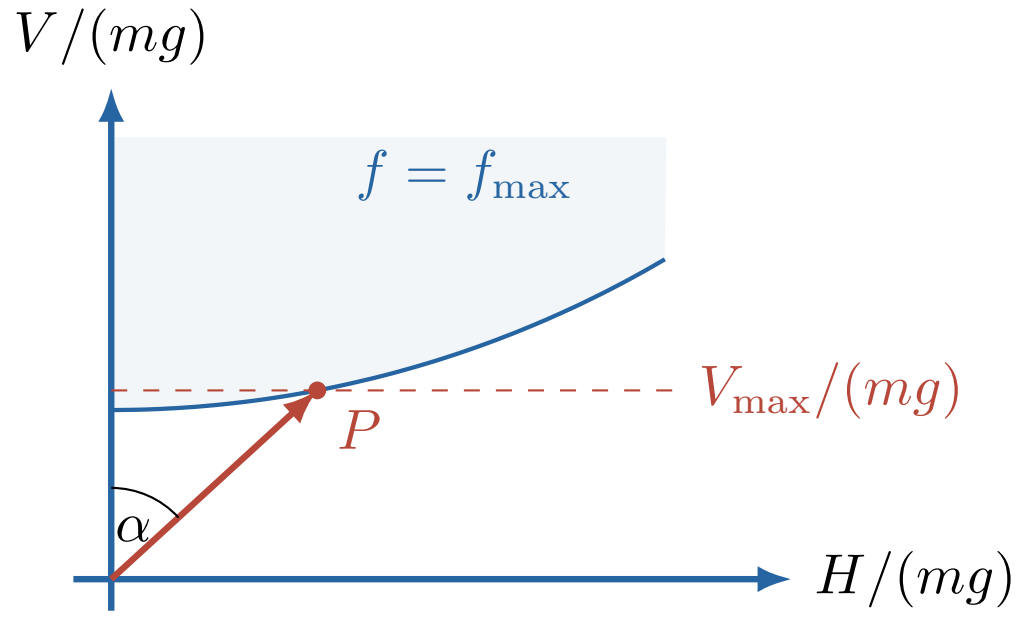


单环平面几何

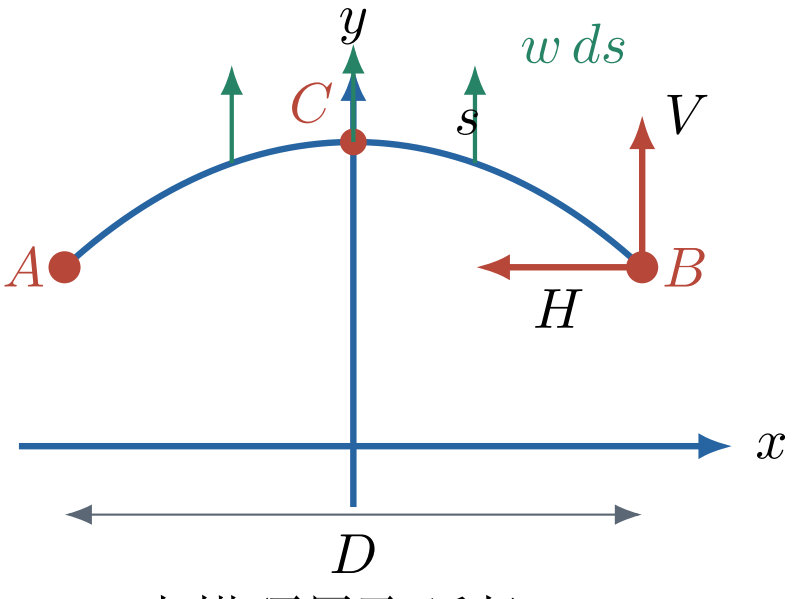


$\rho = s\cos\alpha,\ z = s\sin\alpha$

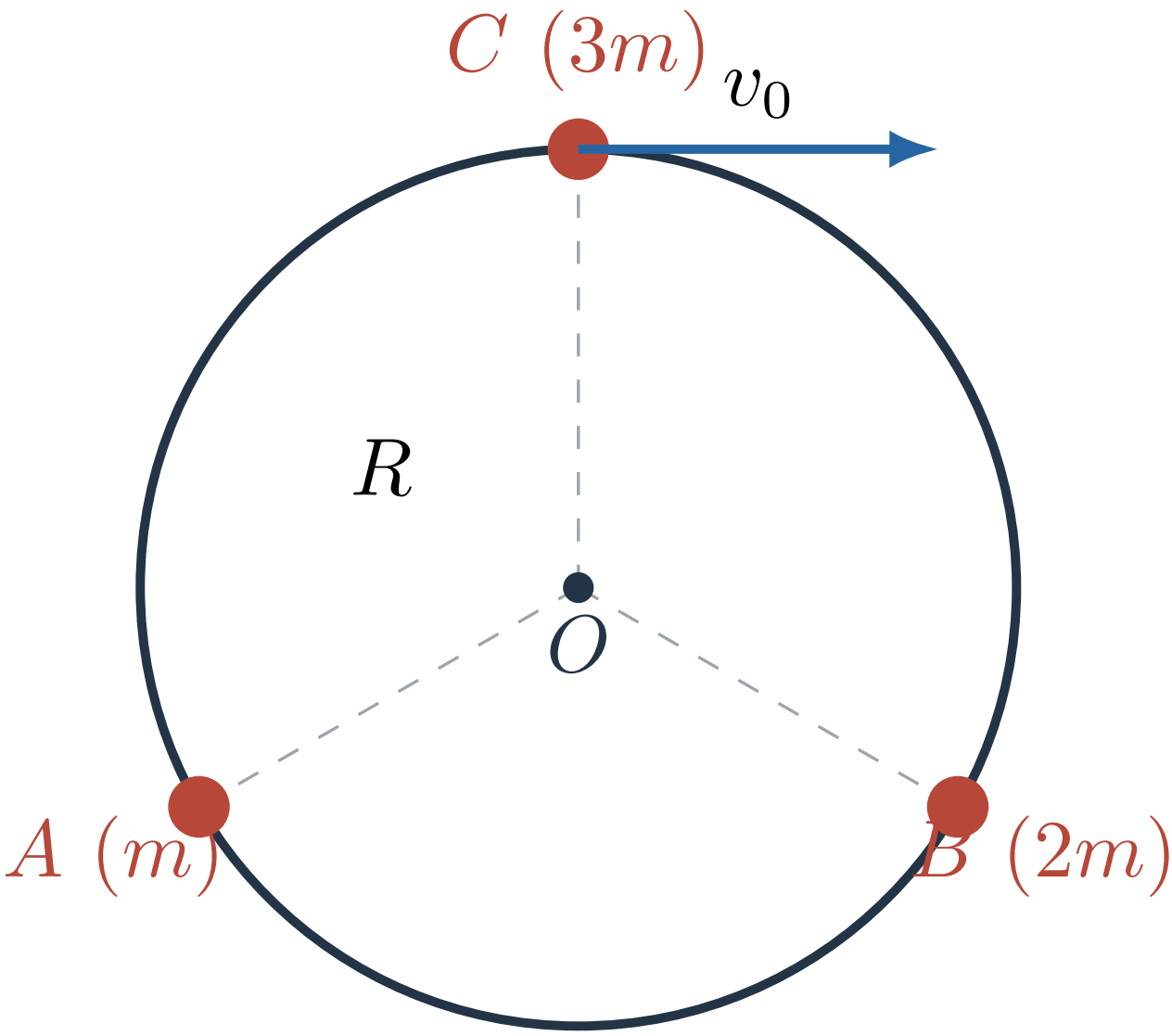

m, l
y
α
α
m
m
θ = 30°
x



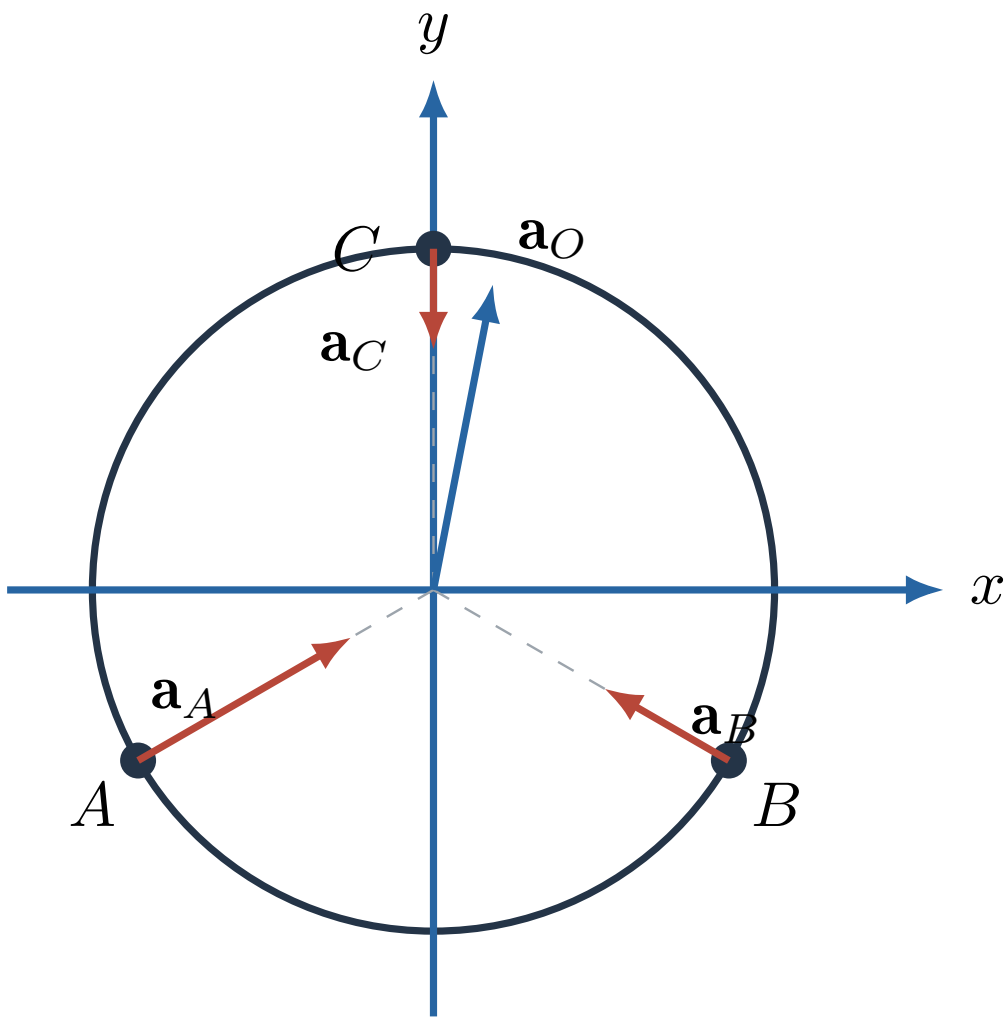
$$H^2 + (mg/2 - V)^2 \leqslant 3m^2g^2/16$$



由拱顶量取弧长 s



水平面内，三个小环相隔 120°



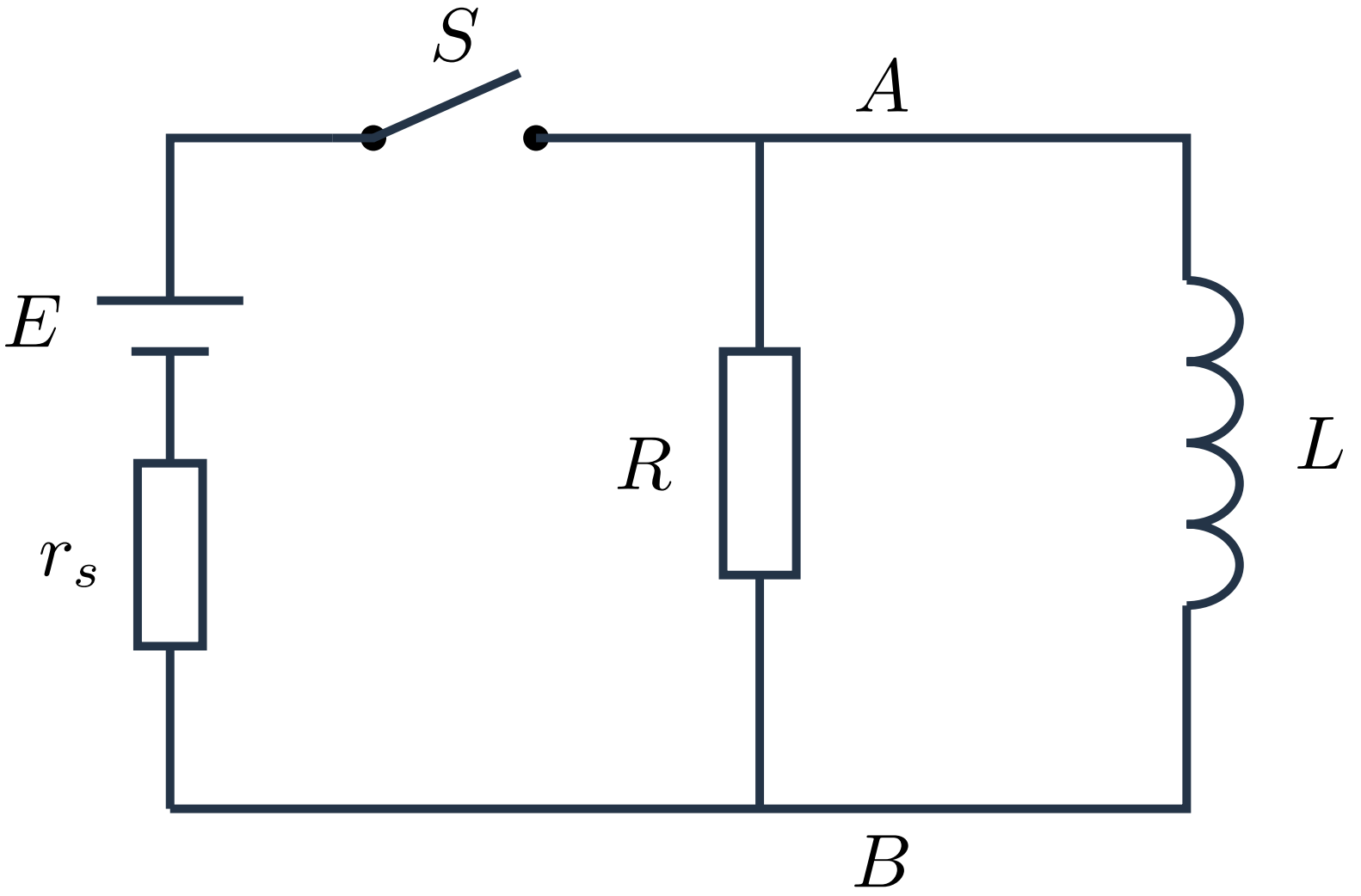
$$\mathbf{a}_O = (a_x, a_y)$$

$$a_y/2 + \sqrt{3}a_x/2 = a_A$$

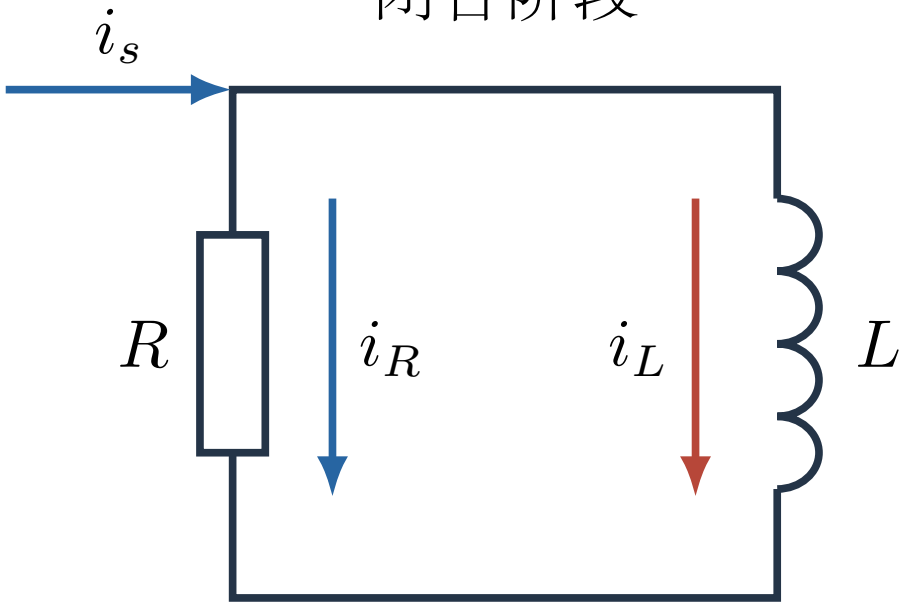
$$a_y/2 - \sqrt{3}a_x/2 = a_B$$

$$a_C + a_y = v_0^2/R$$

红色箭头沿各小环受力方向；蓝色箭头为圆心加速度

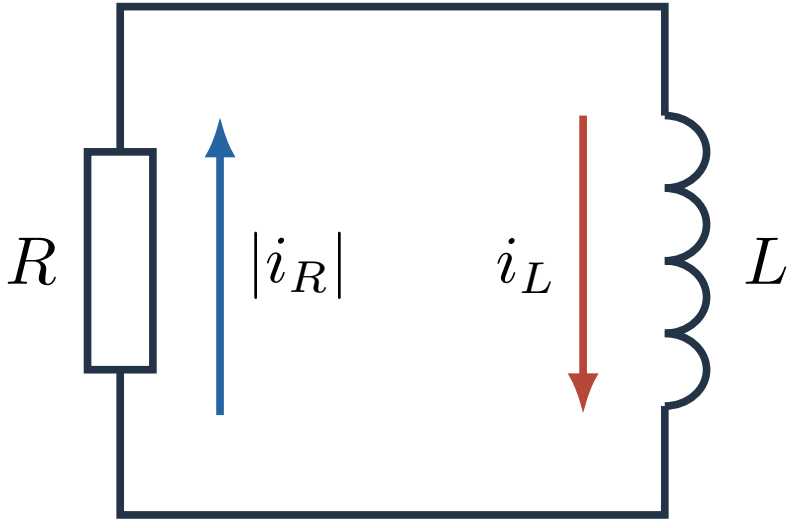

S
A
E
R
L
r_s
B

闭合阶段

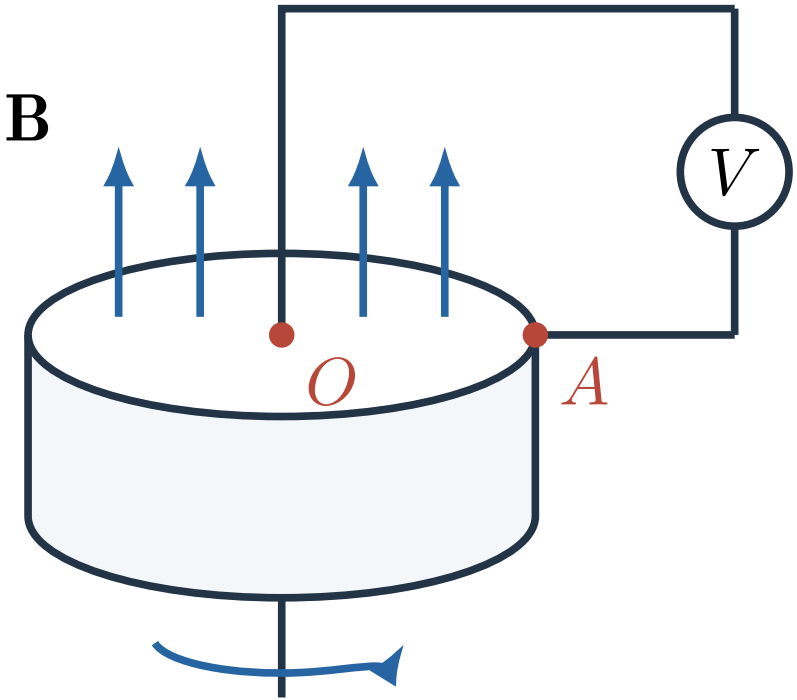


$$Rq_1 = LI_0$$

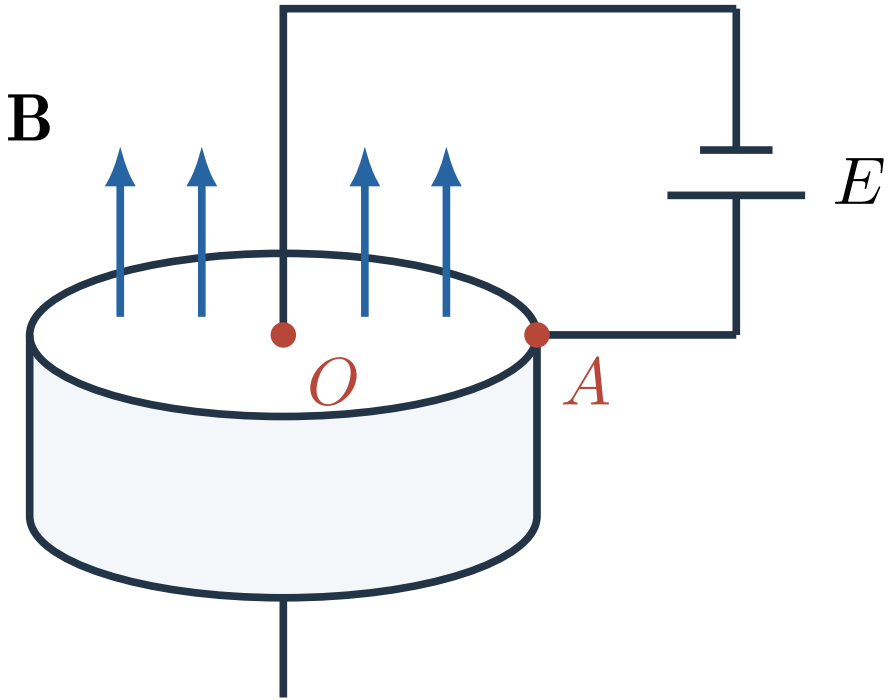
断开阶段



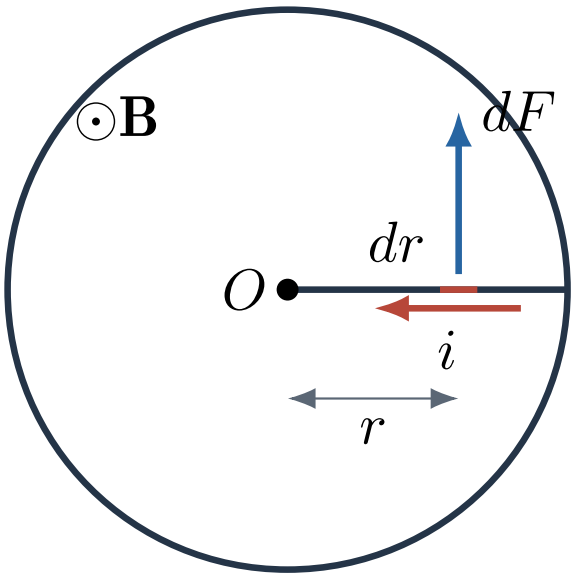
$$Rq_{2,\mathrm{alg}} = -LI_0$$



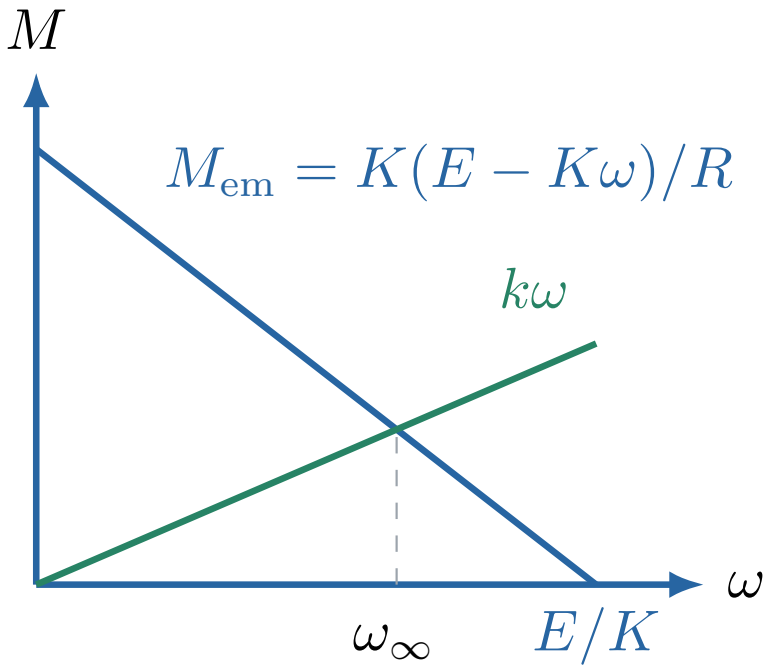
发电测量



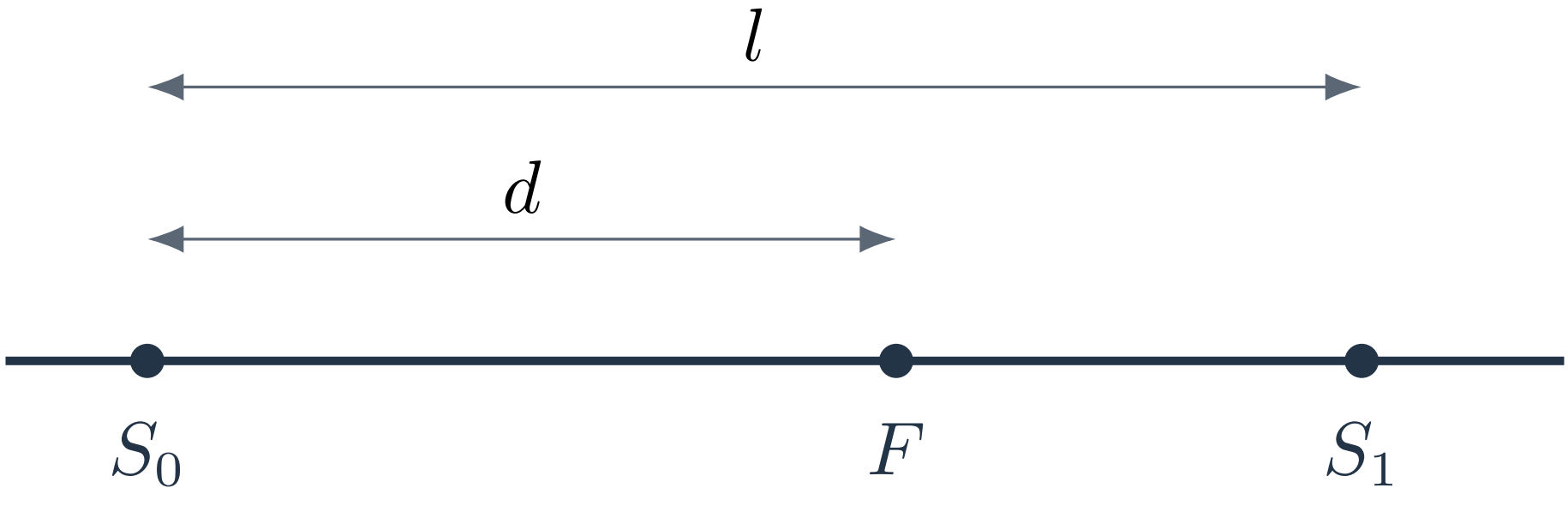
电池下端接圆周

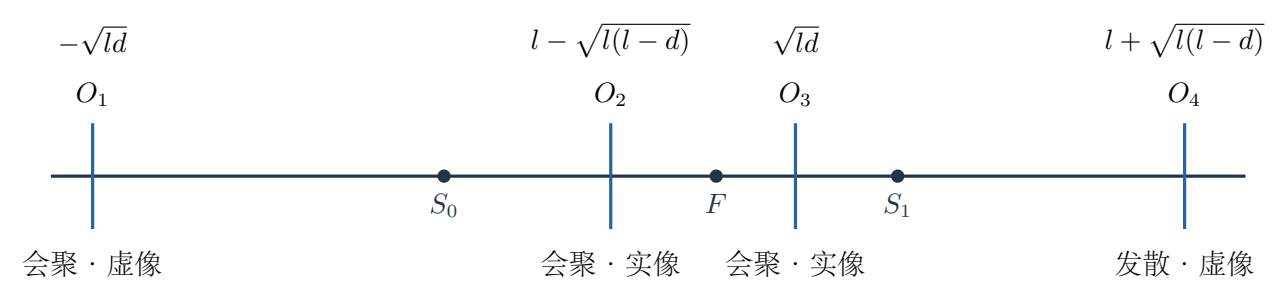


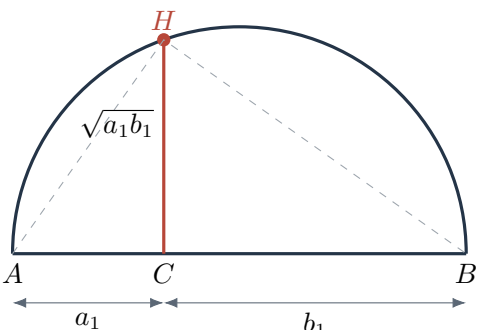
$$dM = r\, iB\, dr$$

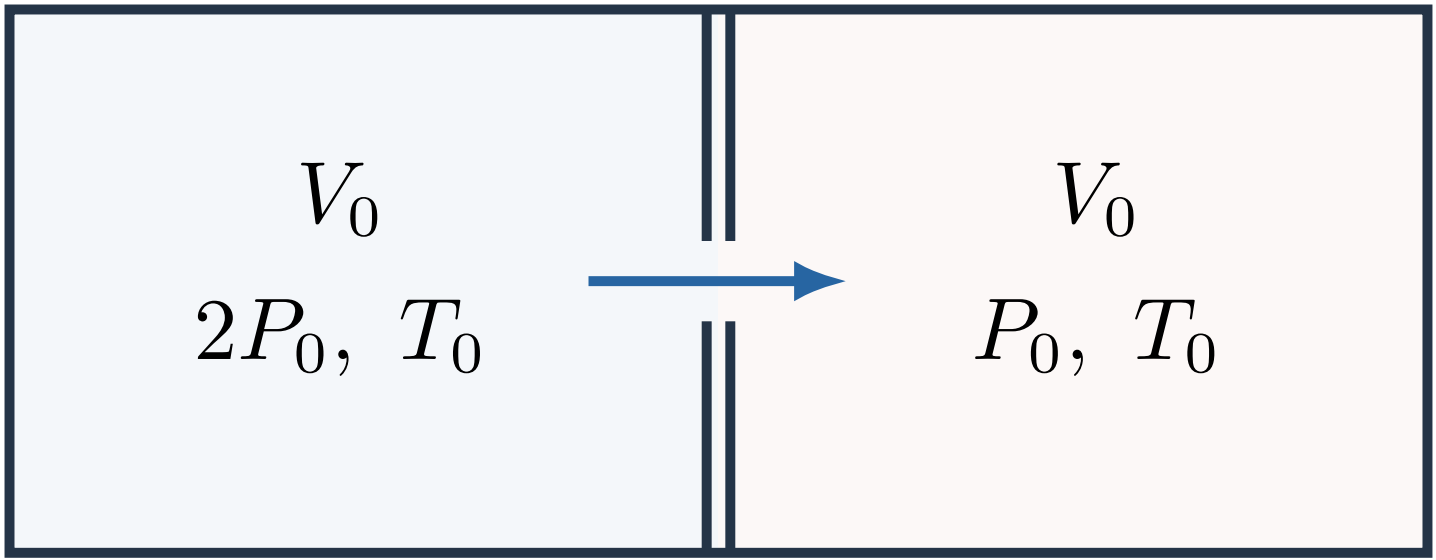


$$K = Br_0^2/2$$


l
d
S_0
F
S_1

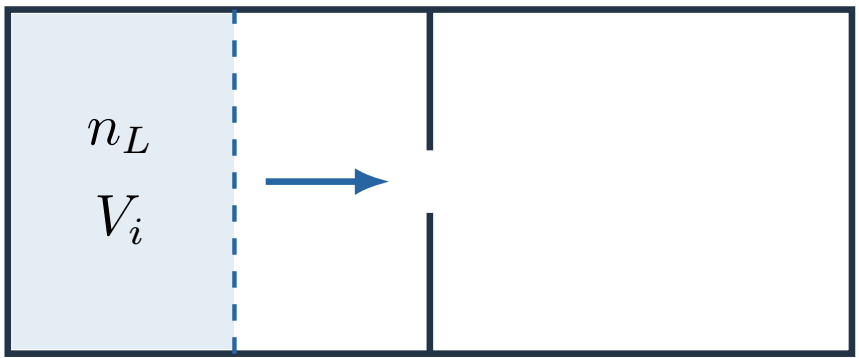

$-\sqrt{ld}$
$l-\sqrt{l(l-d)}$
$\sqrt{ld}$
$l+\sqrt{l(l-d)}$
O_1
O_2
O_3
O_4
S_0
F
S_1
会聚 · 虚像
会聚 · 实像
会聚 · 实像
发散 · 虚像

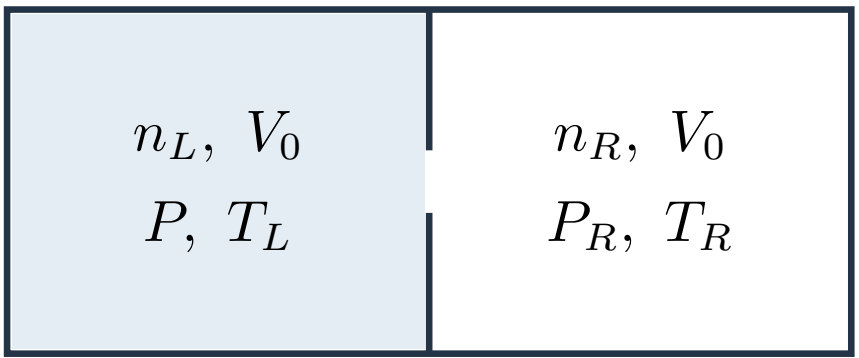

H
$\sqrt{a_1b_1}$
A
C
B
a_1
b_1



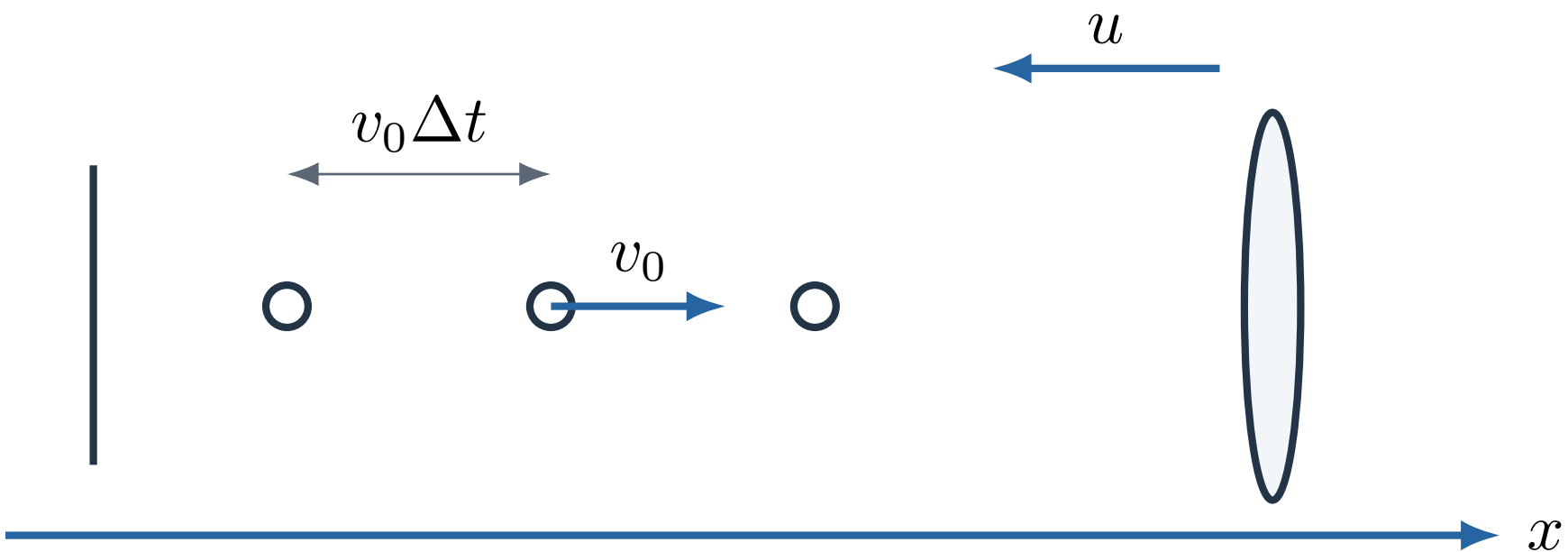
绝热、刚性容器

追踪最终留在左室的同一团气体



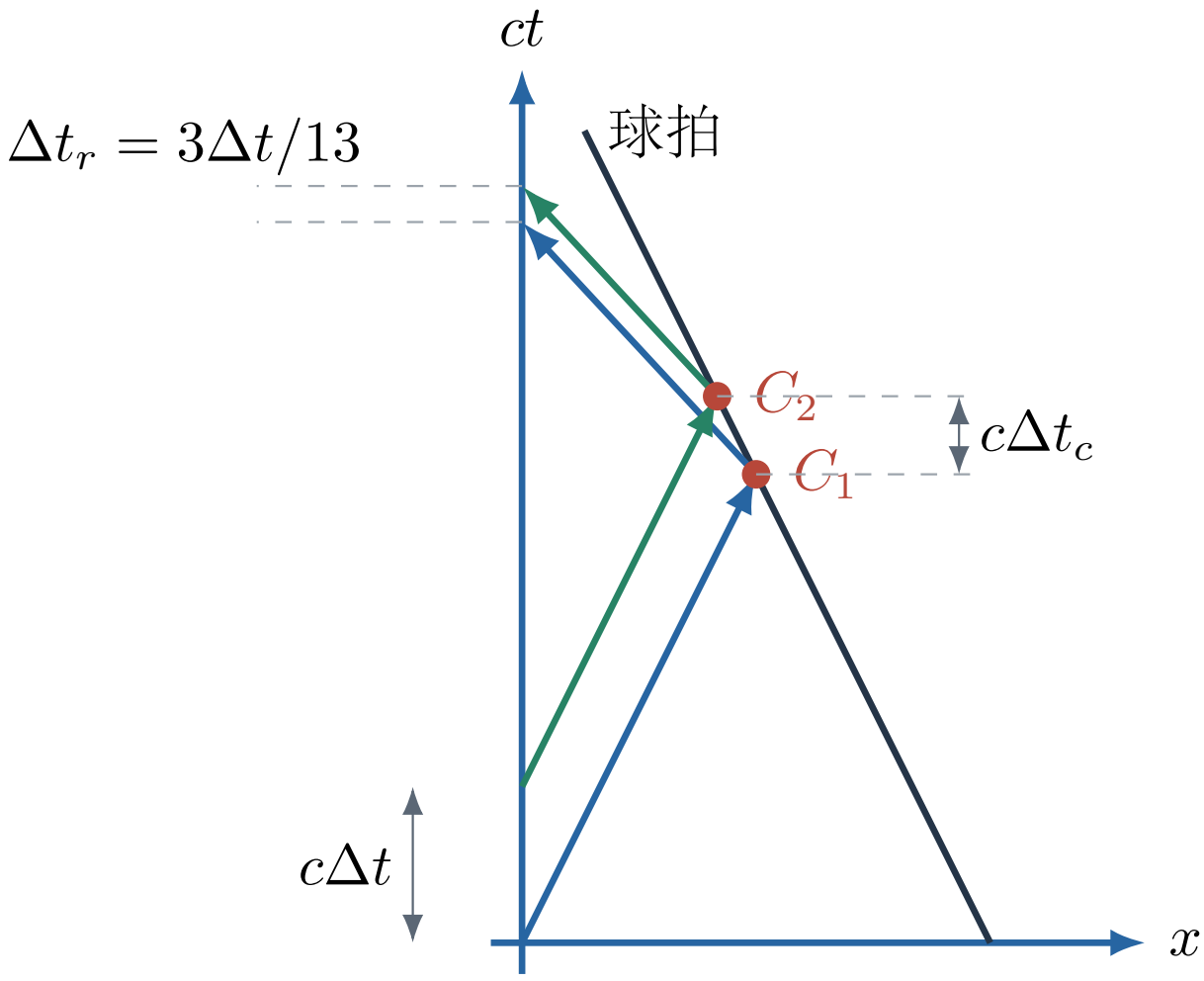


$$2P_0V_i^{\gamma} = PV_0^{\gamma}$$



投球、接球位置　　球拍

$$v_0 = u = c/2$$



两次撞拍：$\Delta t_c = \Delta t/2$

两次回收：$\Delta t_r = 3\Delta t/13$

$\Delta p_{\text{ball}} = p_{\text{out}} - p_{\text{in}}$